



Tutorial 2 – Robot Behaviors and Swarm Systems

Deadline: 31.05.2026

Preparations

In this tutorial we are using a robot simulator. Feel free to use any multi-robot simulator of your choice; a few options, for example, are:

- ▶ Henrik's sim based on Box2D : <https://gitlab.com/InfectedBytes/SwarmRobotics.git>
- ▶ Roborobo : <https://github.com/nekonaute/roborobo3>
- ▶ ARGoS: <http://www.argos-sim.info/>
- ▶ Gazebo: <http://gazebo.org/>
- ▶ Player/Stage: <http://playerstage.sourceforge.net/>
- ▶ JBotEvolver: <https://github.com/BioMachinesLab/jbotevolver>

Try to make a good choice in what is available for your preferred OS and also in terms of overhead created by setting it up.

1 Behaviors of a Single Robot

- ▶ Program a collision avoidance behavior — a robot that moves around while avoiding crashing into walls.
- ▶ Program a wall follower — a robot that moves along the walls without touching them.
- ▶ Program an appropriate behavior for a vacuum cleaning robot. What are strategies to ensure that (almost) everything gets clean?
- ▶ Just play around a little and see what kind of behaviors you can generate.

2 Behaviors of Robot Swarms

- ▶ Program a behavior that makes robots stop if they get close to another robot.
- ▶ Extend your program to limit the time a robot stays stopped according to a defined waiting time.
- ▶ Tweak your implemented behavior and parameters such that your robot swarm eventually aggregates in one big cluster.
- ▶ Analyze how changing the swarm size affects the speed and stability of cluster formation.

Submission Requirements:

- ▶ Groups can consist of a maximum of 2 students
- ▶ Submit one ZIP file named:
`CollRob_02_YOURLASTNAME1_YOURLASTNAME2.zip`
- ▶ Include only necessary files (no large or irrelevant files)
- ▶ Ensure the project is **well-structured and clean**
- ▶ Include all source code, runnable without complex setup
- ▶ Provide a short README explaining how to run the code
- ▶ Include plots/screenshots for each implemented behavior
- ▶ Include a report (implementation, observations, group members, and completed tasks)
- ▶ Include a short video demonstration of the behaviors (.mp4 preferred)
- ▶ Do not include the original task sheet in the submission